ODD Design for the Opinion Leader Influence Model

Feiyu Tang

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Contents

1. Introduction and Purpose

This document describes the Opinion Leader Influence Model (OLIM) using the ODD protocol for agent-based and other simulation models [13]. OLIM is an agent-based model of opinion dynamics on user-generated content and social-media platforms. It represents users as agents who hold continuous latent opinions, express discrete public stances through posts, receive content through a directed follower network and out-of-network exposure, update their opinions after exposure, and revise their follow relations over repeated rounds.

The purpose of OLIM is to study how opinion leaders shape opinion evolution, content circulation, exposure, network adaptation, polarization, homophily, and echo-chamber-like structure. Opinion leaders are not modeled as a separate ontological type of actor. They are ordinary user agents with a leader indicator that gives them structural visibility advantages in several parts of the model: initialization, posting origination, out-of-network diffusion, exposure weight, response coefficients, and creator evaluation.

The model has three conceptual foundations. First, it follows the Continuous Opinions and Discrete Actions (CODA) logic by distinguishing between a user's latent continuous opinion and the discrete stance visible to others through public expression [16]. Second, it draws on Social Judgment Theory, in which responses to messages depend on whether the message is interpreted as aligned, acceptable disagreement, or rejectable disagreement relative to the receiver's current position [19, 14]. Third, it treats online opinion dynamics as a platform-mediated process in which visibility, content style, selective exposure, and adaptive network ties jointly shape aggregate outcomes [12, 1, 18, 4, 10, 6].

OLIM is used to compare no-leader and leader conditions and to evaluate how macro-level outcomes change as leader share, leader opinion direction, initial network topology, population size, and simulation length vary. The implemented experiment workflow includes a core leader-effects matrix, a no-leader control scenario, and a smaller robustness scenario that varies the number of rounds.

The model is abstract and theory-driven. It is not calibrated to a single empirical platform and does not use platform-specific behavioral traces as runtime input. However, an archived Zhihu economic discussion thread is used outside the simulation workflow to ground the network-initialization assumptions and to provide a scenario-level consistency check for leader-direction effects [20]. Parameter values operationalize modeling assumptions about visibility, exposure, expression, response, and tie adaptation. Simulation results should therefore be interpreted as outcomes of a controlled computational model rather than as direct forecasts for a specific platform.

2. Entities, State Variables, and Scales

2.1 Entities

OLIM contains one decision-making entity type: user agents. Each user represents an account or individual participant on a social-media platform. Users can become potential originators, create posts, receive posts, update latent opinions, and revise outgoing follow ties.

Posts are not autonomous agents. They are transient interactional objects created by users within a round. A post carries observable signals: creator identity, expressed stance, expressive style, and whether the creator is an opinion leader. Posts affect the system through diffusion, exposure aggregation, opinion updating, and network adaptation, but they do not persist as independent actors after the round is closed.

The interaction environment is a directed follower network. A directed edge from viewer j to creator i means that j follows i. The in-degree of i is therefore the number of followers attached to i, and is used as a measure of follower-based visibility.

2.2 User State Variables

Code variable	Notation	Description
`node`	i	Unique user identifier.
`o_t`	o_i^t	Latent continuous opinion on the focal bipolar issue, clipped to $[-1, 1]$.
`s_t`	s_i^t	Expressed stance used for interpretation and later action, with values in $\{-1, +1\}$.
`tau_t`	τ_i^t	Confidence or opinion certainty, clipped to $[0.1, \tau_{\max}]$.
`Abar`	$\bar{A}_i$	Baseline activity propensity.
`A_t`	A_i^t	Current activity level, clipped to $[0, 1]$.
`F_t`	F_i^t	Current follower count, equal to current in-degree.
`L`	L_i	Opinion-leader indicator, where 1 denotes leader status.
`O_t`	O_i^t	Potential-originator indicator for the current round.
`C_t`	C_i^t	Actual-creator indicator for the current round.
`M_pC_prev`	$M_{i,+C}^{t-1}$	Previous-round weighted exposure to positive constructive content.
`M_pT_prev`	$M_{i,+T}^{t-1}$	Previous-round weighted exposure to positive toxic content.
`M_nC_prev`	$M_{i,-C}^{t-1}$	Previous-round weighted exposure to negative constructive content.
`M_nT_prev`	$M_{i,-T}^{t-1}$	Previous-round weighted exposure to negative toxic content.

Code variable	Notation	Description
M_pC_t	$M^t_{i,+C}$	Current-round weighted exposure to positive constructive content.
M_pT_t	$M^t_{i,+T}$	Current-round weighted exposure to positive toxic content.
M_nC_t	$M^t_{i,-C}$	Current-round weighted exposure to negative constructive content.
M_nT_t	$M^t_{i,-T}$	Current-round weighted exposure to negative toxic content.

2.3 Post Variables

If user i creates a post in round t, the post is represented as

$$p^t_i = (i, x^t_i, y^t_i, q_i),$$

where $x^t_i \in \{-1, +1\}$ is the expressed stance, $y^t_i \in \{C, T\}$ is the expressive style, and $q_i = L_i$ records whether the creator is an opinion leader. The four observable content categories are positive constructive, positive toxic, negative constructive, and negative toxic.

2.4 Temporal and Spatial Scales

The model runs in discrete rounds. Each round contains posting origination, content creation, diffusion, opinion updating, network adaptation, and state synchronization. Posts exist only within the round in which they are created. Their effects are carried forward indirectly through updated opinions, activity, exposure memory, and network ties.

The model has no explicit geographic space. The relevant space is the directed follower network plus probabilistic out-of-network exposure. Network ties are dynamic because ordinary non-creator viewers can maintain, remove, or form follow relations after encountering creators.

3. Process Overview and Scheduling

OLIM proceeds in synchronous discrete rounds. The implementation schedule is defined in `main.py`. One round consists of the following ordered processes.

1. **Round preparation.** The model copies the current directed graph, resets `O_t` $= 0$ and `C_t` $= 0$, and recomputes each user's follower count F^t_i from current in-degree.

2. **Posting origination.** Each user receives a probability of becoming a potential originator, based on current activity, leader status, and follower count. The potential-originator state O^t_i is sampled from this probability.

3. **Content creation.** Potential originators pass through a round-level creation gate. Those who pass evaluate positive and negative stance utilities. If expression is sufficiently feasible and unambiguous, a post is created with a discrete stance and constructive or toxic style.

4. **Diffusion and exposure.** Created posts are delivered deterministically to followers and probabilistically to non-followers. Ordinary-user posts use a fixed out-of-network probability, while leader posts use a follower-count and style-dependent sigmoid probability. If a viewer receives more posts than the maximum reading capacity, retained posts are sorted by creator follower count. Exposures are aggregated into four weighted content categories.

5. **Opinion updating.** Each user remaps current exposure into aligned and opposite categories relative to its current expressed stance, updates confidence, updates latent opinion, and samples the next expressed stance.

6. **Network adaptation.** Ordinary users who did not create content in the current round evaluate encountered creators and probabilistically maintain, remove, or form follow ties. Opinion leaders and current-round creators are excluded from this adaptation step.

7. **Round closure.** The model updates follower counts, activity, latent opinions, expressed stances, previous exposure memory, graph node attributes, and round-level summary metrics. The next round starts from these synchronized states.

The implementation excludes a creator from receiving its own post. It does not globally exclude all creators from exposure to other creators' posts during the diffusion step. However, agents who created content in the current round are excluded from network adaptation in that round.

4. Design Concepts

4.1 Basic Principles

OLIM is based on three principles. First, it separates latent opinions from public expressions following the CODA framework [16]. Second, it treats message effects as interpretive: the same post may be reinforcing, moderating, or backfiring depending on the viewer's current stance and the post's style [19, 14, 2]. Third, it treats opinion leadership as structural visibility rather than a separate behavioral species [12, 22, 1]. Leader status modifies visibility, weight, and response parameters but does not create a separate class of agent actions.

4.2 Emergence

The model is designed to generate macro-level patterns from local posting, exposure, updating, and tie adaptation. The main emergent outcomes are polarization, opinion extremity, homophily, sign-based modularity, echo-chamber-like clustering, content imbalance, and changes in network density [7, 9, 11, 15].

4.3 Adaptation

Users adapt through several mechanisms. Latent opinions and confidence change after exposure. Activity changes at round closure as a function of baseline activity, recent creation, received exposure, and lack of exposure. Ordinary non-creator viewers revise outgoing follow ties toward encountered creators. Previous-round exposure also affects later content creation through stance and style feedback terms.

4.4 Objectives

Agents do not maximize a global objective. Content creation and network adaptation use local utility-like rules. Stance expression depends on opinion, confidence, prior exposure, posting costs, and stochastic shocks. Tie adaptation depends on local evaluations of encountered creators, including alignment, toxicity, opposition, and leader status.

4.5 Learning

The model contains no explicit learning algorithm. Agents do not estimate model parameters or learn strategies. Nevertheless, their later behavior changes because opinions, confidence, activity, follower counts, network ties, and previous exposure records are updated over time.

4.6 Prediction

Agents do not make forward-looking strategic predictions about future popularity, future opinions, or future network states. All behavioral rules use current state variables, previous exposure memory, and current-round observed content.

4.7 Sensing

Agents observe posts, creator identity, expressed stance, expressive style, leader flag, and whether they currently follow the creator. They do not observe other users' latent opinions. In network adaptation, creator evaluation is therefore based on observed content rather than hidden beliefs.

4.8 Interaction

Interaction is mediated by posts and follow ties. Follow ties determine deterministic follower exposure. Out-of-network exposure represents platform-like visibility beyond existing ties [4, 10]. Exposed posts affect opinion updating, and encountered creators can become targets of tie maintenance, unfollowing, or new following [6].

4.9 Stochasticity

Stochastic processes occur in graph generation, random graph orientation, leader selection where applicable, initial opinions, confidence, activity, initial stance sampling, origination, the creation gate, stance and style utility shocks, stance and style sampling, out-of-network exposure, and network tie updates. Reproducibility is controlled through explicit random seeds.

4.10 Collectives

The model has no formal collective agents. Communities, polarized groups, echo-chamber-like structures, and leader-centered attention patterns are emergent properties of the evolving opinion distribution and directed follower network.

4.11 Observation

The model records round-level summaries of content production, exposure, opinions, and network structure. It also records final-state metrics and can optionally record agent-level opinion trajectories. The observation layer does not feed back into agent decisions except through state variables already defined in the model.

5. Initialization

Initialization is controlled by the model parameter settings. Unless overridden, the model uses $N = 1000$ agents and a Barabasi–Albert network with $m = 3$. The experiment workflow also uses $N \in \{500, 1000, 1500\}$.

5.1 Network Initialization

The model first creates an undirected graph and then randomly orients each undirected edge to form a directed follower network. If the undirected edge connects u and v, one of $u \to v$ or $v \to u$ is selected with equal probability. In the resulting directed graph, an edge from viewer to creator means that the viewer follows the creator.

The implementation supports four topology families:

- **BA:** a Barabasi–Albert graph with $m = 3$, implying target average degree near $2m = 6$ [5].
- **ER:** an Erdos–Renyi graph. If no edge probability is supplied, the probability is chosen to target the configured average degree.
- **WS:** a Watts–Strogatz graph with neighborhood size $k = 6$ and rewiring probability 0.10 [21].
- **SBM:** a stochastic block model with four approximately balanced blocks and 70% of expected edges placed within blocks.

This sparse directed-network design is empirically grounded with an archived Zhihu like-author network from a December 2024 economic discussion thread [20]. In that empirical network, a directed edge from a liker to an answer author is treated as a proxy for attention or exposure. The network contains 20,934 users and 45,616 directed edges, with density 0.000104 and a largest weakly connected component of 20,787 users. Visibility is highly concentrated among answer authors: the top 1% of authors by in-degree account for 35.4% of all author in-degree, the top 5% account for 66.5%, and the top 10% account for 78.1%. These empirical values do not calibrate graph parameters directly, but they support the use of sparse networks with a high-visibility tail and motivate selecting opinion leaders from high in-degree users.

5.2 Leader Initialization

After the directed network is created, each user's follower count F_i^0 is computed from in-degree. Opinion leaders are then assigned using one of two main rules. Under threshold selection, users whose follower count exceeds the configured in-degree threshold receive $L_i = 1$. Under fixed-share selection, the model selects either a random set of users or the highest in-degree users, depending on the configured leader-selection rule. The main experiment matrix uses a fixed leader share and top-in-degree selection. The no-leader control sets the leader share to zero and disables leader selection.

5.3 Opinion, Confidence, Stance, and Activity

Ordinary users' initial latent opinions are drawn from a normal distribution centered near neutrality:

$$o_i^0 \sim \mathrm{clip}_{[-1,1]}\left(\mathcal{N}(\mu_o, \sigma_o)\right),$$

with default mean $\mu_o = 0.0$ and standard deviation $\sigma_o = 0.5$. Opinion leaders, when present, are assigned stronger opinion magnitudes:

$$o_i^0 = d_i r_i, \qquad r_i \sim U(0.6, 0.9),$$

where d_i is controlled by the leader-orientation condition. In the positive condition $d_i = +1$; in the negative condition $d_i = -1$; in the balanced condition d_i is sampled from $\{-1, +1\}$.

Initial confidence for ordinary users is drawn from a normal distribution with configured mean and standard deviation. Leaders use a separate leader-confidence distribution. Confidence is clipped to $[0.1, \tau_{\max}]$. Initial expressed stance is sampled from latent opinion and confidence; leaders' initial expressed stance is set to the sign of their latent opinion.

Baseline activity $\bar{A}_i$ is sampled uniformly from the configured activity range, and the initial current activity is $A_i^0 = \bar{A}_i$. Potential-originator and creator indicators are initialized to zero. Previous exposure memories are initialized to zero for all four content categories. Reproducibility is controlled through explicit random seeds.

6. Input Data

The current implementation uses no external empirical input data during simulation. Agents, networks, opinions, confidence levels, activity propensities, leader status, exposure events, and tie updates are generated from parameter dictionaries and random seeds.

External literature motivates the model structure and the interpretation of mechanisms such as CODA-style expression, social-judgment response, opinion leadership, toxic expression, and adaptive exposure [16, 19, 8, 12, 3, 17, 4]. In addition, the thesis uses an archived Zhihu economic discussion thread as empirical grounding and validation material outside the model runtime [20]. In ODD terms, the Zhihu material is therefore not input data for the simulation; it is external empirical evidence used to document and interpret the model.

6.1 External Zhihu Material

The empirical material comes from a Zhihu question-and-answer thread crawled in December 2024:

https://www.zhihu.com/question/668753879/answer/3649938067

The data-processing workflow and derived empirical artifacts are maintained in a companion repository [20]. The repository is used to document how the archived thread was converted into two thesis-facing forms of evidence: a like-author attention network and answer-level stance balance measures. These materials remain outside the OLIM runtime workflow and are not read by the simulation code.

6.2 Use in the ODD Documentation

The Zhihu material is used in two limited ways. First, the like-author attention network supports the model-design claim that online discussion networks can be sparse and highly concentrated among visible users. This evidence is used to justify the representation of opinion leaders as structurally visible agents, not to initialize a real Zhihu network inside OLIM.

Second, answer-level stance balances are used for scenario-consistency validation. They compare the empirical all-answer direction and high-visibility author direction with existing ABM experiment summaries. This comparison is a post-simulation interpretation step; it does not initialize agents, set parameters, fit coefficients, or feed back into the model schedule.

Simulation outputs can be saved to CSV, JSON, and figure files. These outputs are analysis products rather than model inputs.

7. Submodels

7.1 Posting Origination

At the beginning of each round, every user receives a probability of becoming a potential originator. The implementation computes

$$z_i^t = \alpha_0 + \alpha_1 A_i^t + \alpha_2 L_i + \alpha_3 \log(1 + F_i^t),$$

then maps this value through the sigmoid function and applies an upper cap:

$$\pi_i^t = \min\{\sigma(z_i^t), \pi_{\max}\}.$$

The potential-originator state is sampled as

$$O_i^t \sim \text{Bernoulli}(\pi_i^t).$$

Origination means that the user has an opportunity to create content. It does not guarantee that a post will be produced.

7.2 Content Creation

Only users with $O_i^t = 1$ enter the content-creation stage. The first gate is a round-level creation probability:

$$P^{\text{gate},t} = \texttt{base\_create\_prob} \cdot W_{\text{create}}^t \cdot \exp[-\lambda_{\text{att}}(t-1)],$$

where W_{create}^t is a warmup scale and λ_{att} is `attention_decay`. The same gate probability applies to all selected originators in the round.

If the user passes this gate, the model computes positive and negative stance utilities. Previous exposure memories are first transformed using $\log(1+M)$. Positive expression is favored by prior positive constructive and toxic exposure and disfavored by prior negative constructive and toxic exposure; negative expression is defined symmetrically. In compact form:

$$U_{i,+}^t = \alpha_B \tau_i^t \max(o_i^t, 0) + W_{\text{stance}}^t \Phi_+(\mathbf{M}_i^{t-1}) - c_0 + \varepsilon_i^t,$$

$$U_{i,-}^t = \alpha_B \tau_i^t \max(-o_i^t, 0) + W_{\text{stance}}^t \Phi_-(\mathbf{M}_i^{t-1}) - c_0 + \varepsilon_i^t.$$

Here Φ_+ and Φ_- denote the implementation's weighted previous-exposure feedback terms, W_{stance}^t is the stance-feedback warmup scale, and ε_i^t is a normal shock.

The implementation suppresses posting if both stance utilities are negative:

$$\max(U_{i,+}^t, U_{i,-}^t) < 0.$$

It also suppresses posting if both utilities are non-negative but too similar:

$$U_{i,+}^t \geq 0, \quad U_{i,-}^t \geq 0, \quad |U_{i,+}^t - U_{i,-}^t| \leq \epsilon_{\text{amb}}.$$

These are implementation choices that represent weak or ambiguous motivation to speak.

If a post is created, the utility difference is mapped into a positive-stance probability:

$$p_{i,+}^t = \sigma\left(\frac{U_{i,+}^t - U_{i,-}^t}{T_s}\right).$$

The code then applies confidence-dependent Beta-style smoothing. With $\kappa_i^t = \kappa\tau_i^t$,

$$\alpha_i^t = 1 + \kappa_i^t p_{i,+}^t, \qquad \beta_i^t = 1 + \kappa_i^t(1 - p_{i,+}^t),$$

and the final probability of positive stance is

$$\frac{\alpha_i^t}{\alpha_i^t + \beta_i^t}.$$

The expressed post stance x_i^t is sampled from this probability.

After stance is selected, the model chooses expressive style. Constructive and toxic style utilities are computed from stance-specific previous exposure, opinion extremity, posting costs, warmup scaling, and normal shocks:

$$V_{i,C}^t = \gamma_0 + W_{\text{style}}^t \Psi_C(x_i^t, \mathbf{M}_i^{t-1}) - \gamma_4|o_i^t| - c_C + \eta_{i,C}^t,$$

$$V_{i,T}^t = \delta_0 + W_{\text{style}}^t \Psi_T(x_i^t, \mathbf{M}_i^{t-1}) + \delta_3|o_i^t| - c_T + \eta_{i,T}^t.$$

The probability of constructive expression is

$$P(y_i^t = C) = \sigma\left(\frac{V_{i,C}^t - V_{i,T}^t}{T_C}\right).$$

A created post is stored as a dictionary containing creator id, stance x_i^t, style y_i^t, and leader flag $q_i = L_i$.

7.3 Diffusion and Exposure

For each created post, every other user is considered as a possible viewer. If viewer j follows creator i, meaning $g_{ji}^t = 1$, exposure is deterministic:

$$E_{ji}^t = 1.$$

The creator is excluded from receiving its own post.

If j does not follow i, out-of-network exposure is probabilistic. For ordinary creators,

$$P(E_{ji}^t = 1 \mid g_{ji}^t = 0, L_i = 0) = p_O.$$

For opinion leaders,

$$P(E_{ji}^t = 1 \mid g_{ji}^t = 0, L_i = 1) = \sigma\left(\beta_0^d + \beta_1^d \log(1 + F_i^t) + \beta_2^d \mathbb{I}(y_i^t = T)\right).$$

Thus leader out-of-network visibility depends on follower count and whether the post is toxic.

If a viewer receives more posts than `max_read_capacity`, the implementation sorts received posts by creator follower count and retains only the most visible creators' posts. Dropped posts are removed from the exposure matrix.

After truncation, received posts are aggregated into weighted exposure totals:

$$\mathbf{M}_j^t = (M_{j,+C}^t, M_{j,+T}^t, M_{j,-C}^t, M_{j,-T}^t).$$

Leader posts receive weight w_l, and ordinary-user posts receive weight w_o.

7.4 Opinion Updating

Opinion updating uses the current-round weighted exposure vector. The model first remaps positive and negative exposure into aligned and opposite exposure relative to the viewer's current expressed stance s_i^t. If $s_i^t = +1$, positive exposure is aligned; if $s_i^t = -1$, negative exposure is aligned. Each exposure total is transformed using $\log(1 + M)$.

Let $C_{i,A}^t$, $T_{i,A}^t$, $C_{i,O}^t$, and $T_{i,O}^t$ denote aligned constructive, aligned toxic, opposite constructive, and opposite toxic exposure after this remapping and saturation. The response coefficients are ordinary-user coefficients when $L_i = 0$ and leader-specific coefficients when $L_i = 1$. The opinion pressure is

$$\Delta_i^t = s_i^t \left(\omega_{pC}^{L_i} C_{i,A}^t + \omega_{pT}^{L_i} T_{i,A}^t + \omega_{nC}^{L_i} C_{i,O}^t + \omega_{nT}^{L_i} T_{i,O}^t\right).$$

Aligned exposure generally reinforces the current stance. Opposite constructive exposure can attenuate the current stance when its coefficient is negative, while opposite toxic exposure can produce backfire when its coefficient is positive.

Confidence is updated from the clarity of the received information environment. Let

$$N_{i,+}^t = M_{i,+C}^t + M_{i,+T}^t, \qquad N_{i,-}^t = M_{i,-C}^t + M_{i,-T}^t, \qquad N_i^t = N_{i,+}^t + N_{i,-}^t.$$

The clarity ratio is

$$R_i^t = \begin{cases} |N_{i,+}^t - N_{i,-}^t| / N_i^t, & N_i^t > 0, \\ 0, & N_i^t = 0. \end{cases}$$

Environmental evidence is $\tau^t_{\mathrm{env},i} = \tau_0 \log(1 + N^t_i)$, and confidence becomes

$$\tau^{t+1}_i = \mathrm{clip}_{[0.1,\tau_{\max}]}\left[\tau^t_i + \tau^t_{\mathrm{env},i}(R^t_i - \theta_{\mathrm{conf}})\right].$$

The relative weight of external evidence is

$$W^t_i = \frac{\tau^t_{\mathrm{env},i}}{\tau^t_i + \tau^t_{\mathrm{env},i}},$$

with zero weight when there is no environmental evidence. The latent opinion update is

$$o^{t+1}_i = \mathrm{clip}_{[-1,1]}\left[o^t_i + \left(1 - \frac{1}{2}|o^t_i|\right) W^t_i \Delta^t_i\right].$$

The damping term reduces movement for agents already near the opinion extremes. After updating latent opinion and confidence, the next expressed stance is sampled from the updated values using `sample_stance_from_opinion`.

7.5 Network Adaptation

Network adaptation applies only to ordinary users with $L_j = 0$ who did not create content in the current round. For each such viewer j, the implementation identifies the unique creators encountered in the viewer's exposure set. Each encountered creator is evaluated from the observed post.

Let $A^t_{ji} = \mathbb{I}(x^t_i = s^{t+1}_j)$ denote alignment with the viewer's updated stance and $O^t_{ji} = \mathbb{I}(x^t_i \neq s^{t+1}_j)$ denote opposition. The perceived value term is

$$V^t_{ji} = a_1\mathbb{I}(A^t_{ji}, y^t_i = C) + a_2\mathbb{I}(A^t_{ji}, y^t_i = T) - a_3\mathbb{I}(O^t_{ji}, y^t_i = C) - a_4\mathbb{I}(O^t_{ji}, y^t_i = T).$$

The perceived cost term is

$$K^t_{ji} = b_T\mathbb{I}(y^t_i = T) + b_O O^t_{ji} + b_{TO}\mathbb{I}(O^t_{ji}, y^t_i = T).$$

The base score is

$$S^t_{ji} = \lambda_V V^t_{ji} - \lambda_K K^t_{ji} + \lambda_L L_i.$$

The implementation further reduces this score for opposite content, applies an additional penalty for opposite-toxic content, and applies a disconnect bias when opposite content comes from an already-followed creator.

If viewer j already follows creator i, the tie is maintained with probability

$$P(g^{t+1}_{ji} = 1 \mid g^t_{ji} = 1) = \sigma(\widetilde{S}^t_{ji}),$$

where $\widetilde{S}^t_{ji}$ is the adjusted score. If j does not follow i, a new follow tie is formed with probability

$$P(g_{ji}^{t+1} = 1 \mid g_{ji}^t = 0) = \sigma(\widetilde{S}^t_{ji} - \theta_F).$$

7.6 Round Closure

At round closure, the implementation computes new follower counts from the updated graph. Activity is updated as

$$A_i^{t+1} = \mathrm{clip}_{[0,1]}\left[(1-\rho_A)A_i^t + \rho_A \bar{A}_i + \omega_1 C_i^t + \omega_2 \frac{|\mathcal{E}_i^t|}{N_E} - \omega_3 \mathbb{I}(|\mathcal{E}_i^t| = 0)\right],$$

where $\mathcal{E}_i^t$ is the viewer's retained exposure set. The model then synchronizes o_i^{t+1}, s_i^{t+1}, A_i^{t+1}, and F_i^{t+1} into the current state variables for the next round. Current exposure totals are copied into the previous-exposure memory fields. Graph node attributes for opinion and leader status are updated. Finally, the model records round-level metrics for content production, exposure, opinions, and network structure.

8. Observation and Experiments

8.1 Recorded Outputs

Each simulation run records its parameter settings, seed, initialization metadata, initial and final network states, final agents, round history, final-state summary, posts by round, exposure sets by round, and optional agent-level opinion trajectories. The central run-level record is built from round-level observations. When opinion tracking is enabled, agent-level opinion snapshots are stored by round.

At the end of each round, the model records the number of potential originators, actual creators, constructive posts, toxic posts, support posts, oppose posts, average exposure size, mean opinion, standard deviation of opinion, mean absolute opinion, opinion variance, extremist ratio, moderate ratio, homophily ratio, sign modularity, and edge count. Final-state summaries include final mean opinion, final standard deviation, final mean absolute opinion, final opinion variance, final extremist ratio, final moderate ratio, final homophily ratio, final cross-cutting ratio, final sign modularity, and final edge count.

8.2 Experiment Profiles and Scenarios

The experiment workflow uses predefined profiles for repeated simulation. The main profile uses six seeds and 50 rounds. The extended profile uses ten seeds and 50 rounds.

The core leader-effects grid varies population size, topology, leader share, and leader opinion mode:

$$N \in \{500, 1000, 1500\}, \quad \mathcal{G} \in \{\mathrm{BA}, \mathrm{ER}, \mathrm{WS}, \mathrm{SBM}\}, \quad q \in \{0.01, 0.03, 0.05\},$$

with leader opinion mode in $\{\text{balanced}, \text{positive}, \text{negative}\}$. Core conditions run for $T = 50$ rounds under the main profile. Leaders in the core matrix are selected as the highest in-degree users after network initialization.

The no-leader control scenario uses the same population sizes and topology set but sets leader share to zero and disables leader selection. The robustness scenario varies $T_{\text{rounds}} \in \{35, 50, 65\}$ on a reduced subset of conditions, using $N = 1000$, topologies BA and SBM, leader shares 0.01 and 0.05, and the three leader opinion modes, with a capped number of seeds.

8.3 Exported Artifacts

The experiment workflow exports a manifest, experiment grid, raw results, summary results, paper-oriented tables, analysis-ready CSV files, and figures. Raw results contain one row per condition and seed. Summary results aggregate metrics across seeds and include means, standard deviations, standard errors, and confidence intervals. These exports support later analysis and reporting but are not used as runtime input by the model.

8.4 Empirical Consistency Artifacts

The companion Zhihu analysis repository exports empirical artifacts used to interpret the simulation results [20]. The all-answer empirical balance is defined as

$$\text{balance\_share} = \frac{\text{optimistic answers} - \text{pessimistic answers}}{\text{total answers}}.$$

For the archived Zhihu thread, the labeled all-answer balance is -0.657. High-visibility author groups selected by in-degree are also negative, including -0.818 for the top 1% and -0.815 for the top 5%. These empirical targets are compared with ABM summary results after simulations are complete. The comparison is a validation and interpretation step, not a feedback loop into the model.

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